

$$\underline{\underline{\gcd(a, p^n) \neq 1.}}$$

$$\Leftrightarrow p \mid a$$

So $\gcd(a, p) > 1$ if and only if a is a multiple of p .

The divisors of p^n are other powers

$\{1, p, p^2, p^3, \dots, p^n\}$
i.e. $p, p^2, p^3, \dots, p^n$
since p is a prime!

The multiples of p in $1, 2, 3, \dots, p^n$ are

$$1p, 2p, 3p, 4p, \dots, p^{n-1} \cdot p$$

There are p^{n-1} such multiples.

So there are

$$p^n - p^{n-1}$$

integers in the list that are coprime to p^n .

$$\Rightarrow \phi(p^n) = p^n - p^{n-1} = p^{n-1}(p-1)$$

$$= 2'(2-1) 5'(5-1)$$

$$= 2 \cdot 5 \cdot 4$$

$$= 40.$$

4 (a).

If G is a finite group and H is a subgroup then $|H|$ divides $|G|$.

(b) Let H be a subgroup of a finite group G .

(i) A left coset of H in G is a subset

$$gH = \{ gh : h \in H \}$$

where g is an element of G , called a representative of this left coset.

Similarly, a right coset is defined

$$\text{as } Hg = \{ hg : h \in H \}$$

(ii) let $g_1, g_2 \in G$.

In addition, assume

$$g_1 H \cap g_2 H \neq \emptyset.$$

let $x \in g_1 H \cap g_2 H$.

$$\Rightarrow x = g_1 h_1 = g_2 h_2, \text{ for some } h_1, h_2 \in H.$$

$$\Rightarrow g_1 = g_2 h_2 h_1^{-1}$$

$$\text{let } y \in g_1 H. \text{ and } g_2 = g_1 h_1 h_2^{-1}$$

$$\text{ie. } y = g_1 k, \text{ for some } k \in H.$$

$$= g_2 \underbrace{h_2 h_1^{-1} k}_{\in H}, \text{ from above.}$$

$$\Rightarrow y \in g_2 H, \text{ since } h_2 h_1^{-1} k \in H \text{ as all three factors are in } H \text{ and } H \text{ is a subgroup.}$$

Secondly, let $z \in g_2 H$.

$$\Rightarrow z = g_2 l, \text{ for some } l \in H$$

$$= g_1 \underbrace{h_1 h_2^{-1} l}_{\in H}, \text{ from above.}$$

$$\Rightarrow z \in g_1 H, \text{ since } h_1 h_2^{-1} l \in H.$$

$$\begin{cases} A \text{ or } B \\ \Leftrightarrow (\neg B) \Rightarrow A \\ \Leftrightarrow (\neg A) \Rightarrow B \end{cases}$$

$$|G| = \sum_{\substack{\text{different} \\ \text{cosets } gH}} |gH|$$

$$= \sum_{\substack{\text{different} \\ \text{cosets } gH}} |H|$$

$$= [G:H] |H|$$

where $[G:H] = \text{index of } H \text{ in } G.$

$\therefore$ number of left cosets of H in G

This shows that $|H|$ divides $|G|$, as required.

(c)

$$D_8 = \langle r, s \mid r^8 = e, s^2 = e, sr = r^{-1}s \rangle$$

$$= \{ \textcircled{e}, r, r^2, \dots, r^7, s, rs, r^2s, \dots, r^7s \}$$

$$|D_8| = 16$$

Consider the subgroup H of D_8

consider $C(g) = \{h \in G : hg = gh\}$

Assume that $\langle g \rangle$ is normal in G .

$$\langle g \rangle = \{g^k : k \in \mathbb{Z}\}.$$

$$\Leftrightarrow \forall x \in G \quad x \langle g \rangle x^{-1} \subseteq \langle g \rangle$$

$$\Leftrightarrow \forall x \in G \quad \forall l \in \mathbb{Z} \quad x g^l x^{-1} = g^k \text{ for some } k \in \mathbb{Z}.$$

$$\forall x \in G \quad \forall l \in \mathbb{Z} \quad \exists k \in \mathbb{Z} \quad x g^l x^{-1} = g^k \quad (*)$$

Claim: $C(g)$ is normal in G .

$$\Leftrightarrow \forall x \in G \quad x C(g) x^{-1} \subseteq C(g)$$

~~Let $x \in G$~~

Let $x \in G$, and $y \in C(g)$

and consider $x y x^{-1}$

we need to show $x y x^{-1} \in C(g)$.

$$\text{i.e. } x y x^{-1} g = g x y x^{-1}$$

$$\begin{aligned}
 \underbrace{x y x^{-1}} g &= x y g^k x^{-1}, \text{ by } (*) \\
 &= x \underbrace{g^k} y x^{-1}, \text{ since } y \in C(g) \\
 &= g \underbrace{x y x^{-1}}, \text{ by } (*) \text{ again.} \\
 &\quad \downarrow x^{-1} g = g^k x^{-1} \\
 &\quad \Leftrightarrow g = x g^k x^{-1} \\
 &\quad \Leftrightarrow g x = x g^k
 \end{aligned}$$

Therefore $C(g)$ is normal in G ,
as required.

(d) Explore the condition-

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

